Experiment 5

Log Analysis and Incident Identification

2.1 What is Log Analysis?

Log analysis means reading recorded system events to understand what happened, when it happened, who was involved, and whether the activity is normal or suspicious. A single line may look harmless, but several related lines can reveal a complete incident sequence.

2.2 Important Authentication Evidence

Item	Easy explanation
Timestamp	When the event occurred. Use it to place events in order.
Hostname	Which machine produced the event.
Service and PID	For example sshd[2290], meaning the SSH service process.
Username	The account that was targeted or used.
Source IP	Where the login request came from.
Result	Failed, accepted, session opened, or session closed.

2.3 What Counts as an Anomaly?

- Many failed attempts from one IP address.
- Attempts using usernames that do not exist.
- A successful login immediately after repeated failures.
- Successful access outside expected working hours.
- A new or unexpected source IP address.
- Several usernames tested from the same source IP.

Aim

Analyze Kali Linux authentication logs to identify failed logins, successful access, unusual login times, repeated source IP activity, and correlated security events.

Safety: Use only your own Kali VM with NAT or Host-only networking. Keep all SSH tests on 127.0.0.1. Enter only three wrong passwords manually; do not use brute-force or password-cracking tools.

1. Confirm the system and log source

```
whoami && hostname && date -Is
test -f /var/log/auth.log && echo 'auth.log available' || echo 'Use journalctl'
sudo journalctl -u ssh.service -n 5 --no-pager
```

2. Start SSH and create a temporary user

```
sudo systemctl enable --now ssh
sudo adduser loglab
```

Create a temporary password that is used only for this lab.

3. Generate three failed loginssud

```
ssh loglab@127.0.0.1
```

Enter an incorrect password three times. The connection should end with Permission denied.

4. Find and confirm failed login events

```
sudo grep -Ei 'Failed password|Invalid user|authentication failure' /var/log/auth.log | tail -n 20
```

Cross-check: The same failed attempts should appear in the SSH journal and usually in /var/log/btmp through lastb.

5. if Auth.log does not Exits then journalctl

```
sudo journalctl -u ssh.service --since '-30 min' -o short-iso --no-pager | grep -Ei 'Failed password|Invalid
user|authentication failure'
```

6. check lastb

```
sudo last -Fai | head -n 20
```

No output means no accepted SSH login was found outside the selected working hours.

7. Generate a successful login

```
ssh loglab@127.0.0.1
```

8. Check last

```
last --time-format iso -i | head -n 20
```

9. Find Accepted SSH Logins

```
sudo grep -E 'Accepted (password|publickey)' /var/log/auth.log | tail -n 10
exit
```

10. Check unusual login times

```
sudo journalctl -u ssh.service --since '-7 days' -o short-iso --no-pager | \
awk '/Accepted (password|publickey)/ {
split($1,a,"T");
split(a[2],b,":");
hour=b[1]+0;
if (hour < 8 || hour >= 18) print
}'
```

11. Count failed attempts by IP

```
sudo journalctl -u ssh.service --since '-1 hour' --no-pager | \
grep 'Failed password' | \
awk '{for(i=1;i<=NF;i++) if($i=="from") print $(i+1)}' | \
sort | uniq -c | sort -nr
```

12. Create the SSH timeline

sudo journalctl -u ssh.service --since '-30 min' -o short-iso --no-pager | \

grep -E 'Failed password|Accepted (password|publickey)|session opened|session closed' | \

tee ~/ssh_event_timeline.txt

```
2026-08-31T10:05:21+05:30 kali sshd[2341]: Failed password for loglab from 127.0.0.1 ...
2026-08-31T10:05:26+05:30 kali sshd[2341]: Failed password for loglab from 127.0.0.1 ...
2026-08-31T10:05:31+05:30 kali sshd[2341]: Failed password for loglab from 127.0.0.1 ...
2026-08-31T10:08:12+05:30 kali sshd[2451]: Accepted password for loglab from 127.0.0.1 ...
```

13. Save evidence

sudo journalctl -u ssh.service --since '-1 hour' -o short-iso --no-pager > ~/ssh_lab_evidence.txt

Correlation check	Record from your output
Same username?	_____
Same source IP?	_____
Time between failure and success?	_____
Inside 08:00-18:00?	_____

14. Cleanup

```
sudo deluser --remove-home loglab
# Run the next command only if SSH was enabled solely for this lab:
sudo systemctl disable --now ssh
```

Expected result

- Failed password events are visible in journalctl or auth.log.
- lastb confirms the bad-login records, while last confirms the successful session.
- The timeline connects timestamp, username, source IP, and outcome.
- The evidence file and its SHA-256 hash are created successfully.

Result: The experiment passes when failed and accepted SSH events are identified, correlated, recorded in the worksheet, and preserved with a hash.

Short viva questions

Q. What is /var/log/auth.log?
A. A text log containing authentication and authorization events when rsyslog is configured to write it.

Q. What does lastb read?
A. /var/log/btmp, which stores bad login attempts.

Q. What does last read?
A. /var/log/wtmp, which stores successful login and session records.

Q. Why is the timestamp important?
A. It places events in order and helps decide whether they are part of the same incident.

Q. What is event correlation?
A. Connecting separate log entries using common details such as time, user, IP address, and service.

Q. Does one failed login prove an attack?
A. No. It may be a typing mistake. Repetition, timing, source, and later activity must be considered.

Q. Why is a successful login after failures important?

A. It may indicate that password guessing eventually succeeded, although it can also be a legitimate user correcting a mistake.

Q. Why use journalctl?

A. It reads the systemd journal and can filter by service, time, priority, and other fields.

Q. Why create a hash of evidence?

A. To provide a fingerprint that helps demonstrate whether the saved file changed.